

## Work №4. OUTPUT FEEDBACK MODEL REFERENCE ADAPTIVE CONTROL. AUGMENTED ERROR APPROACH

*Theoretical background.* Consider the plant given by the differential equation (or given by the state-space equation from Work 3)

$$y^{(n)} + a_{n-1}y^{(n-1)} + a_{n-2}y^{(n-2)} + \dots + a_0y = b_mu^m + b_{m-1}u^{m-1} + \dots + b_0u, \quad (4.1)$$

where  $a_i, i = 0, 1, \dots, n-1$ ,  $b_j = 0, 1, \dots, m$  are the constant unknown parameters.

We make the following standard assumptions about the plant.

**Assumption 4.1.** The plant is controllable and observable, i.e., the polynomials  $a(s)$  and  $b(s)$  of the plant transfer function

$$W(s) = \frac{b(s)}{a(s)} = \frac{b_ms^m + b_{m-1}s^{m-1} + \dots + b_0}{s^n + a_{n-1}s^{n-1} + a_{n-2}s^{n-2} + \dots + a_0}$$

have no common roots. ■

**Assumption 4.2.** The polynomial  $b(s)$  is Hurwitz. ■

**Assumption 4.3.** The sign of  $b_m$  is known. Without loss of generality, we will assume  $b_m$  positive. ■

The objective is to design a control law that will compensate the influence of the plant uncertainties on the closed-loop system stability and ensure the equality

$$\lim_{t \rightarrow \infty} (y_M(t) - y(t)) = 0, \quad (4.2)$$

where  $y_M(t)$  is the output of the reference model

$$y_M(t) = \frac{a_{m0}}{s^2 + a_{m1}s + a_{m0}} [g(t)] = W_M(s) [a_{m0}g(t)], \quad (4.3)$$

in which  $g(t)$  is the piece-wise continuous and bounded reference signal,

$$W_M(s) = \frac{a_{m0}}{s^2 + a_{m1}s + a_{m0}}$$

is the reference model transfer function designed using the pole placement (standard polynomials) approach considered in Work 3. We note that the transfer functions  $W(s)$  and  $W_M(s)$  are not SPR. Hence, an augmented error adaptation algorithm should be applied as one of the solutions to this problem.

*Preliminary results.* Together with the plant (4.1), we consider the filters

$$\dot{v}_1 = \Lambda v_1 + \varsigma_{n-1} u, \quad (4.4)$$

$$\dot{v}_2 = \Lambda v_2 + \varsigma_{n-1} y, \quad (4.5)$$

where  $v_1 \in R^{n-1}, v_2 \in R^{n-1}$  are the state vectors,  $\varsigma_{n-1} = [0, 0, \dots, 0, 1]^T$ ,  $\varsigma_{n-1} \in R^{n-1}$ ,

$$\Lambda = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -k_0 & -k_1 & -k_2 & \dots & -k_{n-2} \end{bmatrix}.$$

The matrix  $\Lambda$  is Hurwitz with the companion polynomial

$$K(s) = s^{n-1} + k_{n-2}s^{n-2} + k_{n-3}s^{n-3} + \dots + k_0.$$

It is proved that for any transfer functions  $W(s)$  and  $W_M(s)$  of the relative degree  $n-m$ , there exists a unique vector  $\theta \in R^{2n-1}$  dependent from the unknown parameters of the plant such that the plant (4.1) can be represented in the form

$$y(t) = \frac{b_m}{K_M(s)} \left[ \theta^T \omega(t) + u(t) \right] + \epsilon(t), \quad (4.6)$$

where  $\omega = [v_1^T, v_2^T, y]^T$  is the regressor,  $\epsilon$  is the exponentially decaying term caused by initial conditions.

*Problem solution.* In this work, we represent two solutions (select one).

### **Solution №1.**

According to the first solution, we additionally accept the following assumption.

**Assumption 4.4.** A constant  $b_{\min}$  verifying

$$b_m \geq b_{\min} \quad (4.7)$$

is known. ■

This assumption will be required to avoid division by zero in adjustable control.

Reevaluate the error  $\varepsilon = y_M - y$  in view of (4.3) and (4.6) (in which we omit the exponentially decaying term  $\epsilon(t)$ ) in the following way:

$$\begin{aligned}\varepsilon &= W_M(s)[a_{m0}g(t)] - W_M(s)[\theta^T \omega + b_m u] = W_M(s)[a_{m0}g - \theta^T \omega - b_m u] = \\ &= W_M(s)[a_{m0}g - \theta^T \omega - b_m u]\end{aligned}$$

or

$$\varepsilon = W_M(s)[a_{m0}g - \theta^T \omega - b_m u]. \quad (4.8)$$

The later motivates the design of the adjustable control law

$$u = \frac{1}{\hat{b}_m} \left( -\hat{\theta}^T \omega + a_{m0}g \right), \quad (4.9)$$

where  $\hat{\theta}$  is the vector of adjustable parameters corresponding to the vector  $\theta$ , while  $\hat{b}_m$  is the adjustable parameter corresponding to the scalar  $b_m$ .

**Remark 4.1.** In order to avoid zero crossing for  $\hat{b}_m(t)$  and, as a result, the singularity of the control, it is necessary to use the projection mechanism. Omitting the general description of the projection operator, here we show its particular case. ■

To derive an adaptation algorithm generating  $\hat{\theta}$  and  $\hat{b}_m$ , we replace (4.9) in (4.8) and obtain the error model with measurable output of the form:

$$\varepsilon = W_M(s)[-\tilde{\theta}^T \omega - \tilde{b}_m u] = W_M(s)[\tilde{\mathcal{G}}^T \varphi], \quad (4.10)$$

where  $\tilde{\theta} = \theta - \hat{\theta}$ ,  $\tilde{b}_m = b_m - \hat{b}_m$  are the parametric errors,  $\tilde{\mathcal{G}} = [\tilde{\theta}^T, \tilde{b}_m]^T$ ,

$$\varphi = [-\omega^T, -u]^T.$$

Let us introduce the augmented error

$$\hat{\varepsilon} = \varepsilon - \hat{\mathcal{G}}^T \bar{\varphi} + W_M(s)[\hat{\mathcal{G}}^T \varphi], \quad (4.11)$$

where  $\hat{\mathcal{G}} = [\hat{\theta}^T, \hat{b}_m]^T$  is the aggregated vector of adjustable parameters,

$$\bar{\varphi} = \begin{bmatrix} -\bar{\omega} \\ -\bar{u} \end{bmatrix} = W_M(s)[\varphi], \quad \bar{\omega} = W_M(s)[\omega], \quad \bar{u} = W_M(s)[u].$$

Then, substituting (4.10) into (4.11), we have the static error model

$$\hat{\varepsilon} = \tilde{\mathcal{G}}^T \bar{\varphi}. \quad (4.11)$$

The later motivates the design of the adaptation algorithm

$$\dot{\hat{\mathcal{G}}} = \gamma \Gamma \frac{\bar{\varphi}}{1 + \bar{\omega}^T \bar{\omega}} \hat{\varepsilon}, \quad (4.12)$$

where  $\Gamma$  is the projection operator introduced according to *Remark 4.1* and defined as

$$\Gamma = \begin{cases} I_{2n} & \text{if } \hat{b}_m(t) \geq b_{\min} \\ I_{2n} - \varsigma_{2n} \varsigma_{2n}^T & \text{if } \hat{b}_m(t) < b_{\min} \end{cases} \quad (4.13)$$

$\varsigma_{2n} = [0, 0, \dots, 0, 1]^T$  is the  $2n$ -dimensional coordinate vector. The second row in (4.13) “stops” the function  $\hat{b}_m(t)$  at the instance of crossing the threshold  $b_{\min}$  in order to secure the control (4.9) from division by zero.

**Remark 4.2.** the normalization factor  $1/(1 + \bar{\omega}^T \bar{\omega})$  in (4.12) is used to provide the boundedness of the closed-loop signals in our case when the regressor  $\bar{\omega}$  is not known to be bounded a priori. ■

**Remark 4.3.** from the practical point of view it is reasonable to diversify the rates of parameters tuning for  $\hat{\theta}$  and  $\hat{b}_m$ . To this end, instead of one gain  $\gamma$  we here and hereafter we will use individual gains  $\gamma_1$  and  $\gamma_2$  in the corresponding adaptation algorithms ■

For the sake of simplicity, the algorithm (4.12) can be split into two ones according to the notations and *Remark 4.3*:

$$\dot{\hat{\theta}} = -\gamma_1 \frac{\bar{\omega}}{1 + \bar{\omega}^T \bar{\omega}} \hat{\varepsilon}, \quad (4.14)$$

$$\dot{\hat{b}}_m = \begin{cases} -\gamma_2 \frac{\bar{u}}{1 + \bar{\omega}^T \bar{\omega}} \hat{\varepsilon} & \text{if } \hat{b}_m(t) \geq b_{\min} \\ 0 & \text{if } \hat{b}_m(t) < b_{\min}, \end{cases} \quad (4.15)$$

where  $\gamma_1$  and  $\gamma_2$  are positive gains.

Thus, the adaptive control law consists of the adjustable controller (4.9), the augmented error (4.11), the adaptation algorithm with projection (4.14), (4.15), (4.13), the parameterization filters (4.4), (4.5), and the reference model (4.3) with calculated in Work 3 parameters.

Selecting the Lyapunov function

$$V = \frac{1}{2} \tilde{\mathcal{G}}^T \tilde{\mathcal{G}}$$

and calculating its time derivative in view of (4.12) and (4.13), it is proved that  $\|\tilde{\mathcal{G}}\| \in \mathcal{L}_\infty$ ,  $\hat{\varepsilon}(t), \dot{\hat{\mathcal{G}}}(t) \rightarrow 0$  as  $t \rightarrow \infty$ . Applying to (4.11) the *swapping lemma*, we have

$$\begin{aligned} \hat{\varepsilon} &= \varepsilon - \hat{\mathcal{G}}^T \bar{\varphi} + \frac{1}{K_M(s)} \left[ \hat{\mathcal{G}}^T \bar{\varphi} \right] = \varepsilon - \hat{\mathcal{G}}^T \bar{\varphi} + \hat{\mathcal{G}}^T \bar{\varphi} - \\ &- H_{KC}(s) \left[ H_{KB}(s) \left[ \varphi^T \right] \dot{\hat{\mathcal{G}}} \right] = \varepsilon - H_{KC}(s) \left[ H_{KB}(s) \left[ \varphi^T \right] \dot{\hat{\mathcal{G}}} \right], \end{aligned} \quad (4.16)$$

where  $H_{KC}(s) = c_K^T (Is - A_K)^{-1}$ ,  $H_{KB}(s) = (Is - A_K)^{-1} b_K$  are the transfer function vectors obtained using the triple  $(A_K, b_K, c_K)$  such that

$$\frac{1}{K_M(s)} = c_K^T (Is - A_K)^{-1} b_K.$$

Since  $\hat{\varepsilon}(t), \dot{\hat{\mathcal{G}}}(t) \rightarrow 0$  as  $t \rightarrow \infty$ , while the transfer function  $1/K_M(s)$  is Hurwitz, then from (4.16) we have the convergence  $\varepsilon(t) \rightarrow 0$  as  $t \rightarrow \infty$ .

Therefore, under Assumptions 4.1-4.4 for any initial conditions  $y(0), \dots, y^{(n-1)}(0), \hat{\mathcal{G}}(0)$  the adaptive control ensures the following properties in the closed-loop system:

P4.1.1. all signals in the system are bounded;

P4.1.2.  $\varepsilon(t), \hat{\varepsilon}(t), \dot{\hat{\mathcal{G}}}(t) \rightarrow 0$  as  $t \rightarrow \infty$ ;

P4.1.3. the parametric error  $\tilde{\mathcal{G}}(t)$  tends to zero exponentially fast if the vector  $\bar{\varphi}$  satisfies the persistent excitation condition, i.e., there exist such  $T, \alpha > 0$  that

$$\int_t^{t+T} \bar{\varphi}(\tau) \bar{\varphi}^T(\tau) d\tau > \alpha I$$

P4.1.4. if  $\tilde{\mathcal{G}}(t) \rightarrow 0$  as  $t \rightarrow \infty$ , then there exist optimal gains  $\gamma_1$  and  $\gamma_2$ , for which the rate of convergence  $\tilde{\mathcal{G}}(t) \rightarrow 0$  is maximum.

**Property P4.1.3 depends on the sufficient number of the harmonics in the signal  $g(t)$  that equals the half of dimension of the vector  $\mathcal{G}$ , i.e.,  $n$ .**

### **The order of the work**

1. Using the conditions from Work 3, design a stabilizing adaptive control ( $g(t) \equiv 0$ ). According to the variants,  $m = 0$ ,  $\hat{b}_m \equiv \hat{b}_0$ , and  $b_m \equiv b_0$ .

Make simulation for three different adaptation gains  $\gamma_1$  and fixed  $\gamma_2$ . Make several plots. In the first plot show the regulated variable  $y$ , in the second one show the control  $u$ , in the other plots (two or three, depending on a solution) one show the adjustable parameters.

2. Using the previous stage and initial data given from Work 3, design the tracking adaptive controller. Make simulation for three different adaptation gains  $\gamma_1$  and fixed  $\gamma_2$ .

Make several plots. In the first plot show the control error  $\varepsilon$ , in the second one show the control  $u$ , in the other plots (two or three, depending on a solution) one show the adjustable parameters.

3. Make conclusions after each step of the work.